Squared Codegrees of Expanded Cliques

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Abstract

The expanded triangle $T^{(2k)}$ consists of the three pairwise unions of three disjoint k-sets. We study its Turán density for the sum of squared $(2k-1)$-codegrees, as asked by Balogh, Clemen, and Lidický. In uniformity four, we determine the density to be $9/32$ and prove that every near-extremal hypergraph is close to the odd bipartite construction with part ratio $1:3$. The upper bound is certified by exact rational flag inequalities on seven vertices. Their strict slack outside four parity types yields an elementary quantitative stability proof. For every $k \geq 2$, an unbalanced odd construction gives the lower bound $1/4 + \frac{k-1}{4k}(2k)^{-1/(k-1)}$. A moment argument using the orthogonal decomposition of a symmetric Boolean function proves that the gap between the density and $1/2$ has order $(\log k)/k$. More generally, for every fixed power of two Q, the squared-codegree density of the expanded $(Q+1)$-clique lies below $1-1/Q$ by order $(\log k)/k$. The upper estimate applies to every fixed clique size, while biased binary colorings give the matching order for these binary sizes.

1 Introduction

The sum of squared codegrees distinguishes hypergraphs with the same number of edges by measuring how unevenly their edges are distributed among small vertex sets. For an r-uniform hypergraph H, write $e(H) = |E(H)|$, and for a set $S \subseteq V(H)$ let $d_H(S)$ be the number of edges containing S. Its codegree squared sum is

$$\mathrm{co}_2(H) = \sum_{S \in \binom{V(H)}{r-1}} d_H(S)^2.$$

For a fixed r-graph F, let $\mathrm{exco}_2(n, F)$ be the maximum of $\mathrm{co}_2(H)$ over all F-free r-graphs on n vertices, where freeness refers to copies that need not be induced. Following Balogh, Clemen, and Lidický [1], define

$$\sigma(F) = \lim_{n\to\infty} \frac{\mathrm{exco}_2(n, F)}{\binom{n}{r-1}(n-r+1)^2}.$$

The existence of this limit is recorded in [1]. We often use the homogeneous normalization

$$s_r(H) = \frac{(r-1)!\,\mathrm{co}_2(H)}{n^{r+1}}.$$

Writing $(n)_j = n(n-1)\cdots(n-j+1)$, the ratio of $s_r(H)$ to the preceding normalized codegree sum, whenever nonzero, is $(n)_{r-1}(n-r+1)^2/n^{r+1} = 1 + O_r(1/n)$. The two quantities therefore differ by

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$O_r(1/n)$ uniformly over all n-vertex r-graphs. We write $\pi(F)$ for the ordinary Turán density, the limiting maximum proportion of r-sets that are edges in an F-free r-graph.

For $k \geq 2$, the *expanded triangle* $T^{(2k)}$ has three disjoint vertex sets A, B, C of size k and edge set $\{A\cup B, A\cup C, B\cup C\}$. It is also called the big triangle. Frankl [3] proved that its ordinary Turán density is $1/2$. The standard asymptotic construction partitions the vertices into two equal parts and takes the $2k$-sets meeting either part in an odd number of vertices. Keevash and Sudakov [5, Theorem 1.1] determined the ordinary extremal graphs for sufficiently large orders within this odd bipartite family, allowing unequal part sizes. We optimize the same classical family for the different squared-codegree objective. Balogh, Clemen, and Lidický [1, Section 8.3, Question 8.4] ask for $\sigma(T^{(2k)})$ and discuss this balanced construction as a possible extremal example. Their preceding calculation determines the different objective obtained by squaring the k-codegrees. Throughout this paper, the codegrees being squared have order $2k-1$.

The first result determines the four-uniform case and describes all asymptotically extremal hypergraphs. Given a partition $V = X \sqcup Y$, write $\mathcal{O}_r(X,Y)$ for the r-graph consisting of all r-sets whose intersection with X has odd size. For even r, exchanging X and Y does not change this hypergraph.

Theorem 1.1. *The four-uniform expanded triangle satisfies $\sigma(T^{(4)}) = 9/32$. There is an absolute constant C such that the following holds for every $n \geq 12$ and $\varepsilon \geq 0$. If H is a $T^{(4)}$-free 4-graph on n vertices and $s_4(H) \geq 9/32 - \varepsilon$, then there is a partition $V(H) = X \sqcup Y$ such that*

$$|E(H) \triangle E(\mathcal{O}_4(X,Y))| \leq C(\varepsilon + n^{-1})n^4, \qquad \left|\frac{|X|}{n} - \frac{1}{4}\right| \leq C\sqrt{\varepsilon + n^{-1}}.$$

Consequently, H differs in at most $C\sqrt{\varepsilon + n^{-1}}\, n^4$ edges from an odd construction whose smaller part has size $\lfloor n/4 \rfloor$.

Thus the balanced construction is not extremal for squared triple codegrees. The optimal imbalance follows from a one-variable calculation, but the upper bound requires control of all $T^{(4)}$-free 4-graphs. We obtain that control from ten rational positive semidefinite matrices and an exhaustive check on seven vertices. The matrices, the enumeration, and two independent verification implementations are included in the accompanying `supplement` directory. The proof uses Razborov's flag algebra method [6]. We give its finite sampling formulation, including the signed flag block needed for the exact bound. The certificate has a uniform positive slack on every seven-vertex type that is not odd. A direct reconstruction from three fixed vertices then proves stability without a removal lemma.

For higher even uniformities, the same odd construction gives an explicit improvement over the balanced value $1/4$. All logarithms below are natural.

Theorem 1.2. *For every integer $k \geq 2$,*

$$\sigma(T^{(2k)}) \geq \beta_k := \frac{1}{4} + \frac{k-1}{4k}(2k)^{-1/(k-1)}. \tag{1}$$

Moreover, as $k \to \infty$,

$$\sigma(T^{(2k)}) \leq \frac{1}{2} - \frac{\log k + 1 + \log(4\log 2)}{8k \log 2} + O\left(\frac{(\log k)^2}{k^2}\right). \tag{2}$$

In particular,

$$\frac{1}{8\log 2} \leq \liminf_{k\to\infty} \frac{k(1/2 - \sigma(T^{(2k)}))}{\log k} \leq \limsup_{k\to\infty} \frac{k(1/2 - \sigma(T^{(2k)}))}{\log k} \leq \frac{1}{4}. \tag{3}$$

The general inequality of Brooks and Linz [2, Theorem 5.1], together with $\pi(T^{(2k)}) = 1/2$, gives $\sigma(T^{(2k)}) \le 1/2 - 1/(8k)$. The logarithmic improvement in (2) uses the additional constraint supplied by the triangle-free graph on k-sets. We transfer this middle-order moment constraint to the $(2k-1)$st moment through the orthogonal decomposition of a symmetric function. The lower bound (1) and the upper bound together determine the order of the gap to $1/2$, but not its leading constant. The limit defining $\sigma(T^{(2k)})$ is always taken with k fixed before k tends to infinity.

The moment argument extends to larger expanded cliques. For $s \ge 3$, let $C_s^{(2k)}$ consist of the pairwise unions of s disjoint k-sets; thus $C_3^{(2k)} = T^{(2k)}$. These are the expanded cliques studied by Keevash and Sudakov [5]. For $s = Q+1$ with Q a power of two, Sidorenko's binary construction [7], described in [5, p. 675], gives ordinary density $1 - 1/Q$. We bias its color classes and study its squared codegrees.

Theorem 1.3. *Fix $s \ge 3$ and put $\theta = (s-2)/(s-1)$. As $k \to \infty$,*

$$\sigma(C_s^{(2k)}) \le \theta - \frac{\theta(1-\theta)}{2k \log 2}\left(\log k + 1 + \log \frac{2\log 2}{\theta}\right) + O_s\left(\frac{(\log k)^2}{k^2}\right). \tag{4}$$

If $s = Q+1$ for a power of two $Q \ge 2$, then

$$\sigma(C_{Q+1}^{(2k)}) \ge \theta - \frac{\theta(1-\theta)}{k}(\log k + 1 - \log(1-\theta)) - O_Q\left(\frac{(\log k)^2}{k^2}\right). \tag{5}$$

In particular, $\theta - \sigma(C_{Q+1}^{(2k)}) = \Theta_Q((\log k)/k)$ for every fixed power of two Q. The error terms are independent of the order of the host hypergraph.

The case $Q = 2$ recovers the logarithmic gap in Theorem 1.2. For other clique sizes, we do not claim a matching lower bound.

Section 2 proves Theorems 1.2 and 1.3 and supplies the lower bound for Theorem 1.1. Section 3 proves the four-uniform upper bound and the local estimate used in Section 4 to establish stability. No exact finite extremal formula or finite uniqueness statement is asserted.

2 Constructions and moment bounds

We first optimize the odd construction. We then derive an upper bound from two moments of the edge indicator. The auxiliary graph on k-sets gives the upper estimate for all expanded cliques. Biased binary constructions then prove the matching order in Theorem 1.3.

Proposition 2.1. *Let $k \ge 2$, and let $X_n \sqcup Y_n$ be a partition of n vertices with $|X_n|/n \to x$. The hypergraph $\mathcal{O}_{2k}(X_n, Y_n)$ is $T^{(2k)}$-free, and, with $t = 1 - 2x$,*

$$\lim_{n\to\infty} s_{2k}(\mathcal{O}_{2k}(X_n, Y_n)) = \frac{1+t^2}{4} - \frac{t^{2k}}{2}. \tag{6}$$

The maximum over $0 \le x \le 1$ is β_k, attained precisely when

$$x = \frac{1 \pm (2k)^{-1/(2k-2)}}{2}.$$

Proof. If three disjoint k-sets had all three pairwise unions as edges, the sum of their intersections with X_n, counted over those edges, would be odd. Each vertex is counted twice, so that sum is even. This proves freeness.

Write $a = |X_n|$ and $b = |Y_n|$. A $(2k-1)$-set containing j vertices of X_n has codegree $a-j$ when j is even, and $b-(2k-1-j)$ when j is odd. Consequently its squared-codegree sum is exactly

$$\sum_{\substack{0\le j\le 2k-1\\ j \text{ even}}} \binom{a}{j}\binom{b}{2k-1-j}(a-j)^2 + \sum_{\substack{0\le j\le 2k-1\\ j \text{ odd}}} \binom{a}{j}\binom{b}{2k-1-j}(b-2k+1+j)^2.$$

After normalization and passage to the limit, this becomes

$$x^2\frac{1+(1-2x)^{2k-1}}{2} + (1-x)^2\frac{1-(1-2x)^{2k-1}}{2} = \frac{1+t^2}{4} - \frac{t^{2k}}{2}.$$

For $u = t^2 \in [0,1]$, the derivative of the final expression is $1/4 - ku^{k-1}/2$. It changes sign from positive to negative at $u = (2k)^{-1/(k-1)}$. Substitution gives (1) and the two stated part ratios. □

For $k = 2$, the maximizing ratios are $1:3$ and $3:1$. We record the loss from the optimum in a form that also gives stability. For $u = x(1-x)$,

$$g(x) := 3u - 8u^2 = \frac{9}{32} - 8\left(u - \frac{3}{16}\right)^2. \tag{7}$$

This is the value in (6) when $k = 2$.

The next estimate is the structural input for the higher-uniformity upper bound. Its auxiliary graph is the same graph on k-sets underlying the middle-codegree calculation in [1, Section 8.3].

Lemma 2.2. *Let $s \ge 3$, $k \ge 2$, and let H be a $C_s^{(2k)}$-free $2k$-graph on $n \ge 1$ vertices. Choose independent uniform vertices $X_1, \dots, X_{2k}$, and let $h(X_1, \dots, X_{2k})$ indicate that they are distinct and form an edge. Put $\theta = (s-2)/(s-1)$ and*

$$p = \mathbb{E}h, \qquad M_t = \mathbb{E}\big(\mathbb{E}[h \mid X_1, \dots, X_t]\big)^2 \quad (0 \le t \le 2k).$$

Then $M_k \le \theta p$, $p \le \theta$, and $M_{2k-1} = s_{2k}(H)$.

Proof. We first record a graph inequality. If a K_s-free graph G has N vertices and e edges, then

$$\sum_v d_G(v)^2 \le 2\theta e N.$$

There is nothing to prove when $e = 0$. Otherwise let x be a nonnegative unit Perron eigenvector of its adjacency matrix, with eigenvalue ρ. The weighted form of Turán's theorem, obtained by rational blow-ups and continuity, gives $\sum_{uv\in E(G)} x_u^2 x_v^2 \le \theta/2$. Hence

$$\rho^2 = \left(2\sum_{uv\in E(G)} x_u x_v\right)^2 \le 4e \sum_{uv\in E(G)} x_u^2 x_v^2 \le 2\theta e.$$

The operator norm of the symmetric adjacency matrix equals ρ, so applying it to the all-ones vector proves the degree-square inequality.

Now form a graph on $\binom{V(H)}{k}$, joining disjoint k-sets whose union is an edge. A clique consists of pairwise disjoint blocks, so this graph is K_s-free. It has $N = \binom{n}{k}$ vertices, degrees $d_H(A)$, and $\binom{2k}{k}e(H)/2$ edges. If H has no edges all assertions hold. Otherwise each unordered root and completion has $k!$ orderings, giving

$$M_k = \frac{(k!)^3}{n^{3k}} \sum_{A\in\binom{V(H)}{k}} d_H(A)^2 \le \frac{k!\binom{n}{k}}{n^k}\theta p \le \theta p.$$

Since $M_k \geq p^2$, this also gives $p \leq \theta$. Counting with $2k-1$ ordered roots gives $M_{2k-1} = (2k-1)!\,\mathrm{co}_2(H)/n^{2k+1}$. □

We use the orthogonal decomposition associated with independent coordinates, usually called the Hoeffding decomposition [4]. Its relevant identities are included in the proof.

Lemma 2.3. *Under the hypotheses of Lemma 2.2, suppose $p > 0$. There is a random variable J taking values in $\{1, \dots, 2k\}$ such that, for $m = \mathbb{E}J$,*

$$M_{2k-1} = p - \frac{p(1-p)}{2k} m, \qquad \mathbb{E}a_J \leq \frac{\theta - p}{1-p}, \tag{8}$$

where $a_j = \binom{k}{j}/\binom{2k}{j}$ for $0 \leq j \leq k$ and $a_j = 0$ for $j > k$. The piecewise linear interpolation A of this sequence on $[0, \infty)$ is convex and nonincreasing. In particular,

$$A(m) \leq \frac{\theta - p}{1-p}. \tag{9}$$

Proof. For each $S \subseteq [2k]$, define

$$h_S = \sum_{U \subseteq S} (-1)^{|S|-|U|} \mathbb{E}[h \mid (X_i)_{i \in U}].$$

Inclusion–exclusion gives $h = \sum_S h_S$ and $h_\varnothing = p$. Averaging any coordinate belonging to a nonempty S makes h_S zero. Hence distinct components are orthogonal, and conditioning on the first t coordinates retains exactly those with $S \subseteq [t]$. The symmetry of h makes $\mathbb{E}h_S^2$ depend only on $|S|$. Set $c_j = \sum_{|S|=j} \mathbb{E}h_S^2$. Then

$$c_j \geq 0, \qquad \sum_{j=1}^{2k} c_j = p - p^2, \qquad M_t = p^2 + \sum_{j=1}^{2k} \frac{\binom{t}{j}}{\binom{2k}{j}} c_j.$$

The sum of the c_j is the variance of the Boolean function h. By Lemma 2.2, $0 < p \leq \theta < 1$, so this variance is positive. Define $\mathbb{P}(J = j) = c_j/[p(1-p)]$. The formulas for M_k and M_{2k-1}, together with $M_k \leq \theta p$, give (8).

The sequence a_j is nonincreasing. For $0 \leq j \leq k$, direct cancellation of the binomial coefficients gives

$$a_j - 2a_{j+1} + a_{j+2} = a_j \frac{k(k-1)}{(2k-j)(2k-j-1)} \geq 0.$$

The formula includes $j = k$, with $a_{k+1} = a_{k+2} = 0$, and all subsequent second differences are zero. Thus A is convex. Jensen's inequality gives $A(m) \leq \mathbb{E}a_J$, proving (9). □

Proof of Theorem 1.3. Apply Lemmas 2.2 and 2.3 to a $C_s^{(2k)}$-free hypergraph. The case $p = 0$ is immediate. Otherwise set

$$\epsilon = \theta - p, \qquad v = p(1-p), \qquad D = \theta - M_{2k-1}.$$

Equation (8) gives $D = \epsilon + vm/(2k)$. We bound D uniformly in the number of vertices, with s fixed.

If $D \geq (\log k)/k$, the required estimate holds for large k. We may assume the reverse inequality. Then $\epsilon \leq D$, $v \geq \theta(1-\theta)/2$ for large k, and $m = O_s(\log k)$. Also $\epsilon > 0$: if $\epsilon = 0$, then $\mathbb{E}a_J = 0$, forcing $J \geq k+1$ and $D \geq \theta(1-\theta)(k+1)/(2k)$, a contradiction.

For every integer $0 \le j \le k/2$, the product formula and $\log(1-z) \ge -2z$ for $0 \le z \le 1/2$ give

$$a_j = 2^{-j} \prod_{i=0}^{j-1} \frac{1 - i/k}{1 - i/(2k)} \ge 2^{-j} \exp(-j^2/k).$$

Both integers adjacent to m lie in this range for large k. Linear interpolation and convexity of 2^{-x} imply $A(m) \ge 2^{-m} \exp(-(m+1)^2/k)$. Since $(\theta - p)/(1-p) \le \epsilon/(1-\theta)$, equation (9) yields

$$m \ge \frac{\log((1-\theta)/\epsilon) - (m+1)^2/k}{\log 2}.$$

The same inequality gives $\log((1-\theta)/\epsilon) \le m \log 2 + (m+1)^2/k = O_s(\log k)$. Since $\theta \ge 1/2$, we have $v = \theta(1-\theta) + (2\theta - 1)\epsilon - \epsilon^2 \ge \theta(1-\theta) - \epsilon^2$. Substitution into $D = \epsilon + vm/(2k)$ therefore gives

$$D \ge \epsilon + \frac{\theta(1-\theta)}{2k \log 2} \log \frac{1-\theta}{\epsilon} - O_s\left(\frac{(\log k)^2}{k^2}\right). \tag{10}$$

The interpolation error is $O_s((m+1)^2/k^2)$. The replacement of v costs at most $O_s(\epsilon^2 \log k/k) = O_s((\log k)^3/k^3)$. Thus the displayed error is uniform in both ϵ and the order n of the host hypergraph.

Put $c = \theta(1-\theta)/(2k \log 2)$. The function $z + c \log((1-\theta)/z)$ is minimized over $z > 0$ at $z = c$. Its minimum is $c[\log k + 1 + \log(2 \log 2/\theta)]$. Equation (10) and $M_{2k-1} = s_{2k}(H)$ now give (4) after n tends to infinity with k fixed.

For the lower bound we reweight Sidorenko's construction. Let $Q = 2^d$, $s = Q + 1$, and $\theta = 1 - 1/Q$. Color the vertices by the group $\mathbb{F}_2^d$, with proportions tending to the probability law $\mu = t\delta_0 + (1-t)U$, where $0 \le t \le 1$ and U is uniform on the group. Take a $2k$-set as an edge precisely when its color sum is nonzero. In a forbidden expanded clique the $Q+1$ block sums would be distinct, since two equal sums have a nonedge as their union. The construction is thus $C^{(2k)}_{Q+1}$-free.

Convolution gives $\mu^{*(2k-1)} = t^{2k-1}\delta_0 + (1 - t^{2k-1})U$. For a root with sum z, its limiting normalized codegree is $1 - \mu(z)$. Thus the limiting squared objective is

$$\begin{aligned} \Phi_{Q,k}(t) &= (1 - \theta + \theta t^{2k-1})\theta^2(1-t)^2 + \theta(1 - t^{2k-1})(\theta + (1-\theta)t)^2 \\ &= \theta^2 + \theta(1-\theta)t^2 - \theta t^{2k}(2\theta - (2\theta - 1)t). \end{aligned} \tag{11}$$

All choices of t are realized by integer class sizes with asymptotic proportions μ. Set $L = \log(k/(1-\theta))$ and $t = \exp(-L/(2k))$. Then

$$t^{2k} = \frac{1-\theta}{k}, \qquad 1 - t^2 = \frac{L}{k} + O_Q\left(\frac{(\log k)^2}{k^2}\right), \qquad 2\theta - (2\theta - 1)t = 1 + O_Q\left(\frac{\log k}{k}\right).$$

Subtracting (11) from θ gives (5). Its leading loss and that of (4) are positive constant multiples of $(\log k)/k$, proving the final assertion. □

Proof of Theorem 1.2. Proposition 2.1 gives (1), and (4) with $s = 3$ gives (2). Expanding the explicit lower bound yields

$$\frac{1}{2} - \beta_k = \frac{\log(2k) + 1}{4k} + O\left(\frac{(\log k)^2}{k^2}\right).$$

This and (2) give the outer inequalities in (3); the middle one follows from the definitions of lower and upper limits. □

3 The four-uniform upper bound

For the remainder of the paper, let $T = T^{(4)} = \{1234, 1256, 3456\}$. We describe the exact finite certificate and prove its implication for arbitrary T-free hypergraphs. A *flag* below is an induced configuration together with distinguished root vertices. Its feature vector records the possible configurations of the additional vertices. The products used in the certificate are averages of outer products for two disjoint extensions of the same roots.

We first express the objective on a fixed number of vertices. For a six-vertex 4-graph J, form the ordinary graph G_J by taking the complements of its edges inside $V(J)$. The graph J is T-free if and only if G_J has no perfect matching. Let

$$w(G) = \sum_{v \in V(G)} \binom{d_G(v)}{2}, \qquad f_6(J) = \frac{w(G_J)}{60}.$$

A pair of adjacent edges of G_J corresponds to two edges of J meeting in three vertices. Among the $\binom{6}{3} \cdot 3 \cdot 2 = 120$ choices of a root triple and two ordered distinct extension vertices, each such pair is counted twice. This explains the denominator 60. For a seven-vertex 4-graph K, define $f_7(K) = \frac{1}{7} \sum_{v \in V(K)} f_6(K - v)$. Double counting ordered pairs of distinct extensions of a triple gives, for every n-vertex 4-graph H with $n \geq 7$,

$$\mathbb{E} f_7(H[W]) = \frac{\mathrm{co}_2(H) - 4e(H)}{\binom{n}{3}(n-3)(n-4)}, \tag{12}$$

where W is a uniform seven-set. The subtraction uses $\sum_{S \in \binom{V(H)}{3}} d_H(S) = 4e(H)$.

We use ten feature blocks. Each matrix below is a finite average of an outer product; its entries need not be nonnegative, and the matrix itself need not be positive semidefinite. All bit positions start at zero. When a graph is encoded by an integer, its pairs or four-sets are listed in lexicographic order, with the first set represented by the least significant bit.

The first two blocks use four ordered roots $R = (r_0, r_1, r_2, r_3)$ and one additional vertex a. In set operations, R denotes the underlying set of roots. The root type is $t = \mathbf{1}_{R \in E(H)}$, where $\mathbf{1}$ denotes an indicator. The flag index is

$$i_R(a) = \sum_{j=0}^{3} 2^j \mathbf{1}_{\{a\} \cup (R \setminus \{r_j\}) \in E(H)}.$$

For a seven-vertex host, choose six vertices uniformly and order them as $(a, b, r_0, r_1, r_2, r_3)$. Define M_t to be the average of $e_{i_R(a)} e_{i_R(b)}^{\mathsf{T}}$ when the root type is t, with zero contribution otherwise. Here e_i is a standard unit vector in $\mathbb{R}^{16}$. Thus each block has denominator $7 \cdot 6! = 5040$. In the graph G_J, the type is the bit of ab, and the two flag indices are the four bits of br_j and ar_j, respectively.

The third block uses three ordered roots $R = (r_0, r_1, r_2)$ and an unordered extension $\{a, b\}$. Its index is

$$i_R(a, b) = \sum_{j=0}^{2} 2^j \mathbf{1}_{\{a,b\} \cup (R \setminus \{r_j\}) \in E(H)} + 8\big(\mathbf{1}_{R \cup \{a\} \in E(H)} + \mathbf{1}_{R \cup \{b\} \in E(H)}\big).$$

There are 24 indices. In a seven-vertex host, choose the ordered roots and then a two-set from the four remaining vertices; its complement is the second extension. The average outer product is denoted by M_3. There are $7 \cdot 6 \cdot 5\binom{4}{2} = 1260$ choices.

The next six blocks use five roots and one additional vertex. A 4-graph on five roots is determined up to isomorphism by its number d of edges. Label the roots $0, \ldots, 4$ so that deleting root j leaves

an edge exactly when $j < d$. The group of root automorphisms is $\Gamma_d = S_d \times S_{5-d}$. An extension vertex a defines a graph g on the roots, with $ij \in E(g)$ exactly when $\{a\} \cup (R \setminus \{i, j\})$ is a hyperedge. The extension is T-free precisely when $g - r$ contains no perfect matching for every $r < d$. Indeed, in the six-vertex complement graph, a perfect matching has one edge joining a to such an r, and its other two edges form a perfect matching of $g - r$.

Group the allowed graphs g into Γ_d-orbits, ordered by their smallest integer representatives. The numbers of orbits, for $d = 0, \ldots, 5$, are

$$34, \quad 42, \quad 48, \quad 31, \quad 19, \quad 6.$$

The feature vector is the unit vector for the orbit of g. For each of the 21 five-root sets in a seven-vertex host, average the two orders of the remaining extension vertices. The resulting block $M_{5,d}$ has denominator 42 and contributes zero at roots of a different type. The orbit label is independent of the chosen legal labeling of the roots.

The final block uses the same five-root type with $d = 2$, but retains a signed feature. On $\Gamma_2 = S_{\{0,1\}} \times S_{\{2,3,4\}}$, define

$$\chi(\pi) = \operatorname{sgn}(\pi|_{\{0,1\}}) \operatorname{sgn}(\pi|_{\{2,3,4\}}).$$

Discard any orbit whose stabilizer contains an element with $\chi = -1$. For each remaining orbit, choose its smallest representative g_i and define $\phi(g_i^\pi) = \chi(\pi) e_i$, where g_i^π is the graph obtained by permuting its vertices by π. The stabilizer condition makes this well-defined. Set $\phi = 0$ on the discarded orbits. There are 12 remaining representatives, in increasing order,

$$20, 21, 22, 23, 58, 59, 130, 131, 148, 149, 150, 151.$$

A different legal root labeling multiplies both extension features by the same sign. Their outer product is therefore unchanged. Averaging this outer product over the same 42 choices defines M_χ, with zero contribution for $d \neq 2$. The coordinates of this signed feature are linear combinations of ordinary labeled flag indicators. Its products therefore fit the same positive semidefinite framework.

Let $\mathcal{S}$ be the set of odd seven-vertex 4-graphs. Up to isomorphism, these have smaller part sizes $0, 1, 2, 3$. The finite statement needed below includes a strictly positive gap outside $\mathcal{S}$.

Lemma 3.1 (Rational certificate)**.** *There are rational positive semidefinite matrices Q_b, indexed in order by*

$$b = 0, 1, 3, (5,0), (5,1), \ldots, (5,5), \chi,$$

with dimensions $16, 16, 24, 34, 42, 48, 31, 19, 6, 12$*, such that every T-free seven-vertex 4-graph K satisfies*

$$f_7(K) + \sum_b \langle Q_b, M_b(K) \rangle \leq \frac{9}{32} - \mu \mathbf{1}_{K \notin \mathcal{S}}, \qquad \mu = \frac{135223}{250000000}. \tag{13}$$

Here $\langle A, B \rangle = \sum_{i,j} A_{ij} B_{ij}$ includes both off-diagonal entries. The maximum diagonal entries of the two four-root blocks, the three-root block, each ordinary five-root block, and the signed block are respectively less than 1*,* 4*,* 1*, and* 1*.*

Exact finite verification. The integer matrices specifying the certificate are part of the supplementary material, in `certificate.json`. Each block has the form

$$Q_b = 10^{-8} B_b R_b B_b^{\mathsf{T}}, \tag{14}$$

where B_b and R_b have integer entries and R_b is symmetric. In the order stated in the lemma, the dimensions of R_b are

$$8, 8, 16, 32, 42, 46, 31, 17, 6, 12.$$

Every leading principal minor of every R_b is positive. These assertions are verified by fraction-free integer elimination. More explicitly, starting with $A = R_b$ and previous pivot $p = 1$, at step j the trailing entries are replaced by

$$A_{uv} \longleftarrow \frac{A_{jj}A_{uv} - A_{uj}A_{jv}}{p} \quad (u, v > j), \qquad p \longleftarrow A_{jj},$$

using the entries before that step. Every division is exact, and every pivot is strictly positive. The pivots are the leading principal minors of R_b, so Sylvester's criterion proves positive definiteness. Equation (14) then proves that Q_b is positive semidefinite. The stated diagonal bounds follow by the same integer matrix multiplications.

For completeness, the host enumeration can be checked without an external graph database. Among the 2^{15} ordinary graphs on six labeled vertices, exactly 7945 have no perfect matching. Canonicalization under all 6! permutations gives 55 types. Fix a corresponding six-vertex 4-graph J and add a seventh vertex v. Its possible new edges are indexed by the 20 triples of $V(J)$. Two such triples A, B are incompatible exactly when $|A \cap B| = 1$ and $A \triangle B \in E(J)$. Every copy of T containing v uses exactly two edges through v, so the allowed extensions are precisely the independent sets of this incompatibility graph.

Enumerating all 2^{20} subsets for each of the 55 types gives 1801387 allowed extensions. Factoring by the automorphisms of the fixed six-vertex graph leaves 125940 representatives with the seventh vertex distinguished. Every allowed seven-vertex graph occurs among these representatives. Hosts with identical target and all ten coefficient matrices may be combined; this gives 19894 rows, not a count of unrooted isomorphism types.

All rows are verified over the common denominator $5040 \cdot 10^8$. In this normalization, the four-root counts are summed over the seven deletions, the three-root counts are multiplied by 4, and all five-root counts are multiplied by 120. The target is the sum of the seven wedge counts divided by 420. The least positive numerator in

$$\frac{9}{32} - f_7(K) - \sum_b \langle Q_b, M_b(K) \rangle$$

is 272609568. Dividing by $5040 \cdot 10^8$ gives exactly μ. The zero rows are precisely the four odd types. In lexicographic four-set encoding, one representative for each is

$$0, \quad 34359705600, \quad 7465221528, \quad 26894515608.$$

The minimum positive slack is attained at the host encoded by 16940822528.

The supplementary verifier reconstructs every host and every flag coefficient, checks all these integer inequalities, and tests the positive principal minors. Its independent implementation reconstructs the signed flags from the root group and tests the T-free condition directly on four-edges; it checks all 125940 hosts without merging coefficient rows and verifies the coverage of all 1801387 extensions. Thus neither the enumeration nor the inequalities require a numerical semidefinite solver. The certificate file has SHA-256 digest

`e182640fde2c1396663675bb3f50ad20180f861012f194fb31871c7afda5de43.`

The command `python3 supplement/verify_all.py` performs both checks using Python's standard library and a C++17 compiler. This establishes the finite assertions in the lemma. □

We next justify the averaging step, where positivity enters. It is important that positivity is asserted for independent extensions conditional on their common roots, not for the individual matrices $M_b(K)$ in Lemma 3.1.

Lemma 3.2. *Let H be a T-free 4-graph on $n \geq 7$ vertices, and let $\delta(H)$ be the fraction of its seven-sets inducing a graph outside $\mathcal{S}$. Then*

$$s_4(H) + \mu\delta(H) \leq \frac{9}{32} + E(n) + \frac{1}{n}, \qquad E(n) = \frac{19n - 88}{(n-5)(n-6)}. \tag{15}$$

For $n \geq 12$, the right-hand error $E(n) + 1/n$ is at most $41/n$.

Proof. Fix a root configuration and let Z be its feature vector for a uniform extension. With two independent extensions Z, Z', a positive semidefinite matrix Q satisfies

$$\mathbb{E}[Z^{\mathsf{T}} Q Z'] = (\mathbb{E}Z)^{\mathsf{T}} Q (\mathbb{E}Z) \geq 0.$$

The vectors in use are unit vectors, signed unit vectors, or zero. If K bounds the diagonal entries of Q, then positive semidefiniteness gives $|Q_{ij}| \leq \sqrt{Q_{ii}Q_{jj}} \leq K$. Hence $|Z^{\mathsf{T}} Q Z'| \leq K$. If the independent extensions intersect with probability ρ, conditioning them to be disjoint gives a lower bound $-K\rho/(1-\rho)$ for the average product.

For four roots, the extensions are individual vertices chosen from $n-4$ vertices. Their collision ratio $\rho/(1-\rho)$ is $1/(n-5)$. The two root types are exclusive, and their maximum diagonal is less than 1. Their combined contribution is therefore at least $-1/(n-5)$.

For three roots, choose independent two-sets from the remaining $n-3$ vertices. The probability that they intersect is

$$\rho = 1 - \frac{\binom{n-5}{2}}{\binom{n-3}{2}} = \frac{4n-18}{(n-3)(n-4)}.$$

The diagonal bound 4 gives a contribution at least $-4(4n-18)/[(n-5)(n-6)]$.

For five roots, the extension collision ratio is $1/(n-6)$. Exactly one ordinary root type applies; the corresponding diagonal bound is less than 1. When the root type is 2, the signed feature adds a second block with diagonal bound less than 1. Equivalently, use the direct sum of the two features and matrices. Their combined contribution is at least $-2/(n-6)$. The simultaneous sign change caused by relabeling the roots does not affect this argument.

Averaging over the roots and adding the three estimates gives

$$\mathbb{E}\sum_b \langle Q_b, M_b(H[W])\rangle \geq -\frac{1}{n-5} - \frac{4(4n-18)}{(n-5)(n-6)} - \frac{2}{n-6} = -E(n).$$

Average (13) over uniform seven-sets W. This bounds $\mathbb{E}f_7(H[W]) + \mu\delta(H)$ by $9/32 + E(n)$. By (12),

$$\mathbb{E}f_7(H[W]) \geq \frac{6(\mathrm{co}_2(H) - 4e(H))}{n^5} \geq s_4(H) - \frac{1}{n}.$$

Here $\mathrm{co}_2(H) \geq 4e(H)$ and $24e(H) \leq n^4$. This proves (15). Finally,

$$40(n-5)(n-6) - n(19n-88) = 21n^2 - 352n + 1200 \geq 0 \qquad (n \geq 12),$$

which gives the claimed error bound. □

Proposition 2.1 gives $\sigma(T) \geq 9/32$. Dropping the nonnegative term $\mu\delta(H)$ in (15) and letting n tend to infinity proves the reverse inequality. This establishes the density assertion in Theorem 1.1. The retained slack controls the local errors needed for stability.

4 Stability

The certificate forces a near-extremal hypergraph to induce odd graphs on almost all seven-sets. The following reconstruction lemma turns this local condition into a global partition. It applies to arbitrary 4-graphs, with no forbidden-subgraph assumption.

Lemma 4.1. *Let H be a 4-graph on $n \geq 7$ vertices, and let δ be the fraction of its seven-sets that are not odd. There is a partition $V(H) = X \sqcup Y$ for which*

$$|E(H) \triangle E(\mathcal{O}_4(X,Y))| \leq \delta\binom{n-3}{4} + \binom{n}{4} - \binom{n-3}{4} \leq \left(\delta + \frac{12}{n}\right)\binom{n}{4}. \tag{16}$$

Proof. Choose a uniform three-set R and then a uniform four-set S disjoint from R. Their union is a uniform seven-set. For each $v \notin R$, define a color $c_v \in \mathbb{F}_2$ by $c_v = \mathbf{1}_{R\cup\{v\}\in E(H)}$.

Suppose $H[R \cup S]$ is odd. Choose local colors $b_u \in \mathbb{F}_2$ witnessing this. For $v \in S$,

$$c_v = b_v + \sum_{u\in R} b_u.$$

Summing over the four vertices of S cancels the root contribution. Thus $\mathbf{1}_{S\in E(H)} = \sum_{v\in S} c_v$ in $\mathbb{F}_2$. The probability that this equality fails is at most δ. Averaging over R, one choice of roots has at most $\delta\binom{n-3}{4}$ failures among the four-sets outside R.

Use the colors c_v on $V(H) \setminus R$ and give the three roots arbitrary colors. The corresponding odd graph disagrees with H on at most the preceding number of outside four-sets, plus all four-sets meeting R. This gives the first inequality in (16). The proportion of four-sets meeting R is at most their expected intersection size with R, namely $12/n$, proving the second. □

Proof of the stability assertion in Theorem 1.1. Suppose $s_4(H) \geq 9/32 - \varepsilon$ and $n \geq 12$. Lemma 3.2 gives

$$\delta(H) \leq \frac{\varepsilon + 41/n}{\mu}. \tag{17}$$

Apply Lemma 4.1 and write $B = \mathcal{O}_4(X,Y)$ for the resulting odd graph. Let $L = |E(H) \triangle E(B)|$. Changing a single four-edge changes four triple codegrees by one. The square of each codegree changes by at most $2n$, so

$$|\operatorname{co}_2(H) - \operatorname{co}_2(B)| \leq 8nL, \qquad s_4(B) \geq s_4(H) - \frac{48L}{n^4} \geq s_4(H) - 2\left(\delta(H) + \frac{12}{n}\right).$$

Exchange the parts if necessary and set $x = |X|/n \leq 1/2$ and $u = x(1-x)$. Sampling the vertex colors independently, with repetitions allowed, gives the polynomial $g(x)$ in (7). Requiring distinct vertices in each edge can only delete nonnegative contributions, so the finite graph satisfies $s_4(B) \leq g(x)$. Therefore

$$8\left(u - \frac{3}{16}\right)^2 \leq \varepsilon + 2\delta(H) + \frac{24}{n}.$$

Since $|u - 3/16| = |x - 1/4|(3/4 - x) \geq |x - 1/4|/4$, it follows that

$$\left|x - \frac{1}{4}\right| \leq \sqrt{2\varepsilon + 4\delta(H) + 48/n}.$$

Equations (16) and (17) now give both asserted estimates for the partition.

To obtain a partition whose smaller part has size $\lfloor n/4 \rfloor$, move at most $n|x - 1/4| + 1$ vertices. Changing the color of one vertex changes at most $\binom{n-1}{3}$ edges of an odd graph. Adding these changes to the previous L proves the final assertion when $\varepsilon + 1/n \leq 1$. For larger values, the trivial bound $\binom{n}{4}$ suffices after increasing the absolute constant C. This completes the proof. □

For $k \geq 3$, the exact value of $\sigma(T^{(2k)})$ remains open. In particular, the moment argument does not show that the odd construction is optimal in higher uniformity. The bounds in (3) leave open both the existence and the value of a leading constant for the logarithmic gap. The same questions remain for the binary expanded cliques of Theorem 1.3; for other clique sizes, even a matching-order lower bound is not proved here.

Computational material and data availability. The directory `supplement` contains the exact certificate and all source files needed to reconstruct and verify its finite inequalities. GPT-6.1 (OpenAI) assisted with implementing the verification programs. Their checks use exact integer arithmetic. Its `README.txt` specifies the inputs, command, output, and relation between each verification stage and Lemma 3.1.

Declaration of AI use. The author used GPT-6.1 to assist with exploring proof strategies, developing verification programs, checking calculations, and improving the exposition. The author takes full responsibility for the content of this paper.

References

[1] J. Balogh, F. C. Clemen, and B. Lidický, Hypergraph Turán problems in ℓ_2-norm, in *Surveys in Combinatorics 2022*, A. Nixon and S. Prendiville (eds.), London Math. Soc. Lecture Note Ser. **481**, Cambridge University Press, Cambridge, 2022, pp. 21–63. Updated version arXiv:2108.10406v3 (2025); question numbering follows this version.

[2] G. Brooks and W. Linz, Some exact and asymptotic results for hypergraph Turán problems in ℓ_2-norm, *European J. Combin.* **136** (2026), 104391. Also available as arXiv:2310.09379v3.

[3] P. Frankl, Asymptotic solution of a Turán-type problem, *Graphs Combin.* **6** (1990), 223–227.

[4] W. Hoeffding, A class of statistics with asymptotically normal distribution, *Ann. Math. Statist.* **19** (1948), 293–325.

[5] P. Keevash and B. Sudakov, On a hypergraph Turán problem of Frankl, *Combinatorica* **25** (2005), 673–706.

[6] A. A. Razborov, Flag algebras, *J. Symbolic Logic* **72** (2007), 1239–1282.

[7] A. Sidorenko, An analytic approach to extremal problems for graphs and hypergraphs, in *Extremal problems for finite sets* (Visegrád, 1991), Bolyai Soc. Math. Stud. **3**, János Bolyai Math. Soc., Budapest, 1994, pp. 423–455.